

Assignment for PhD at SUGAR

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1 Deliverable 1

For a perspective camera model, the parameters are:

1.1 Intrinsic Parameters (Matrix K)

These parameters describe the internal characteristics of the camera:

- **Focal Length** (f_x, f_y): Determines the scaling factors in the x and y axes.
- **Principal Point** (c_x, c_y): The coordinates of the image center.
- **Skew Coefficient** (s): Accounts for any skew due to the non-orthogonal nature of the x and y pixel axes (usually 0 for most cameras).

The intrinsic matrix K is represented as:

$$K = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$$

1.2 Extrinsic Parameters

These parameters describe the position and orientation of the camera in the world:

- **Rotation Matrix** (R): Describes the orientation of the camera.
- **Translation Vector** (t): Describes the position of the camera.

1.3 Distortion Parameters

These account for lens-induced distortions:

- **Radial Distortion Coefficients** (k_1, k_2, k_3): Corrects for radial distortion.

$$x_{\text{distorted}} = x \cdot (1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

$$y_{\text{distorted}} = y \cdot (1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

where $r^2 = x^2 + y^2$ and (x, y) are the normalized image coordinates.
Usually, k_3 is only needed in case of severe distortion.

- **Tangential Distortion Coefficients** (p_1, p_2): Corrects for tangential distortion.

$$x_{\text{distorted}} = x + (2p_1xy + p_2(r^2 + 2x^2))$$

$$y_{\text{distorted}} = y + (p_1(r^2 + 2y^2) + 2p_2xy)$$

where (x, y) are the normalized image coordinates.

The vector with all the distortion parameters is given as:

$$\text{Distortion Coefficients} = [k_1, k_2, p_1, p_2, k_3]$$

1.4 Sources

List of some of the source materials I have used to learn more about the topic:

1. *Matlab documentation* [1].
2. *Camera Calibration class from Columbia University* [2]. Complete walk-through, including derivation of the camera model.
3. *CS231A Course Notes 1: Camera Models* [3]. MIT course material.
4. *PhD thesis "Contributions to Plane-based and Deformable 3D Reconstruction from Monocular Images" by Toby Collins* [4]. Just to make sure of the notation used by other PhDs in the group and I also learned more about PSfM.

2 Deliverable 2

I will approach this task in two ways. First, I will answer this by going from the simple pinhole camera model described above using my prior knowledge of what a possible neural network could look like for the prediction and delivering an example code in the next task. After that, I will work on it as I would in the PhD, looking at the literature for state-of-art methods and then suggesting some things we could try based on that.

2.1 First attempt

For this, I will focus on the prediction of intrinsic parameters and distortion coefficients (no extrinsics prediction). I will also simplify the problem by assuming that the skew coefficient (s) is zero.

Data Types

- *Input data*: Single image frames (3D tensor of pixel values, shape [height, width, channels]). We could also have a sequence of frames (4D tensor) as input and add an LSTM layer to capture temporal information, but, to keep things simple, here we are only predicting on single frames.
- *Output data*: Predicted intrinsic parameters and distortion coefficients. Vector of shape: $[f_x, f_y, c_x, c_y, k_1, k_2, p_1, p_2, k_3]$.

Constraints I interpreted this part of this item to mean the physical and mathematical constraints that the model assumes. Below I describe the constraints and how they were implemented in the neural network itself.

- The focal length (f_x, f_y) should be positive. I use the *softplus* activation function to ensure that the focal lengths are always positive.
- The principal point (c_x, c_y) should lie within the image dimensions. I use the *sigmoid* function to constrain c_x and c_y to be within the image boundaries.
- The distortion coefficients $(k_1, k_2, p_1, p_2, k_3)$ can be positive or negative, but they usually fall within certain ranges. I use the *tanh* function to limit the range of distortion coefficients to $[-1, 1]$.

2.2 Second attempt

Conveniently, there is a very recent survey paper [5] about deep learning methods for camera calibration. From the methods presented, I was really interested in the self-calibration method by Hagemann [6].

They introduced a novel approach for intrinsic camera self-calibration that combines deep learning with explicit modeling of projection functions and multi-view geometry. The key component is a self-calibrating bundle adjustment (SC-BA) layer that can be integrated into deep learning architectures. This layer optimizes camera intrinsics along with poses and depth through differentiable Gauss-Newton steps. The authors implement this within the DROID-SLAM framework, resulting in a system called DroidCalib that can infer camera intrinsics from monocular video during operation, without requiring calibration targets.

The SC-BA layer optimizes camera intrinsics jointly with camera poses and depth. This means that as the system improves its estimate of the camera’s intrinsic parameters, it simultaneously refines its depth estimates. While still challenging, depth estimation in laparoscopic videos has seen some recent progress [7,8], indicating that this method could possibly work for our use case.

To adapt DroidCalib for laparoscopic videos, several key modifications would probably be necessary, such as:

- The system would need enhanced robustness to handle dynamic scenes with tissue deformation and instrument movement, possibly incorporating confidence-aware estimation for both feature matches and depth estimates.
- The feature extraction and matching processes would need to be specialized to cope with challenges like specular reflections, smoke, and blood.
- A multi-scale approach might be beneficial to handle varying depths in laparoscopic scenes.
- The unified camera model might need fine-tuning for laparoscopic optics.
- Additional modules for surgical tool detection and stronger temporal consistency checks could improve performance.
- Finally, incorporating scene-specific priors about typical laparoscopic structures could improve estimation in low-texture areas.

In any case, this may be something worth looking into.

3 Deliverable 3

The code for the simple model described in section 2.1 is available in the folder *simple_model* in the GitHub repository.

As a bonus code, I have also attempted to use a real dataset (Standard Pinhole Model dataset [5]) to learn the focal length from images. This was just something I threw together quickly before I worked on the rest of the assignment. I repurposed an old project code that used EfficientNetV2 as a feature extractor and just adapted it to a regression task for the prediction of the focal length. The code for this is in the *pinhole_test* folder.

3.1 Results with Standard Pinhole Model dataset

The model was trained for 50 epochs (1h33 runtime). The best validation Mean Squared Error (MSE) score was at epoch 47, with a value of 7349.98. The test set MSE was 3976.42, which used the weights saved in epoch 47. Figure 1 contains the graphs for the training and validation.

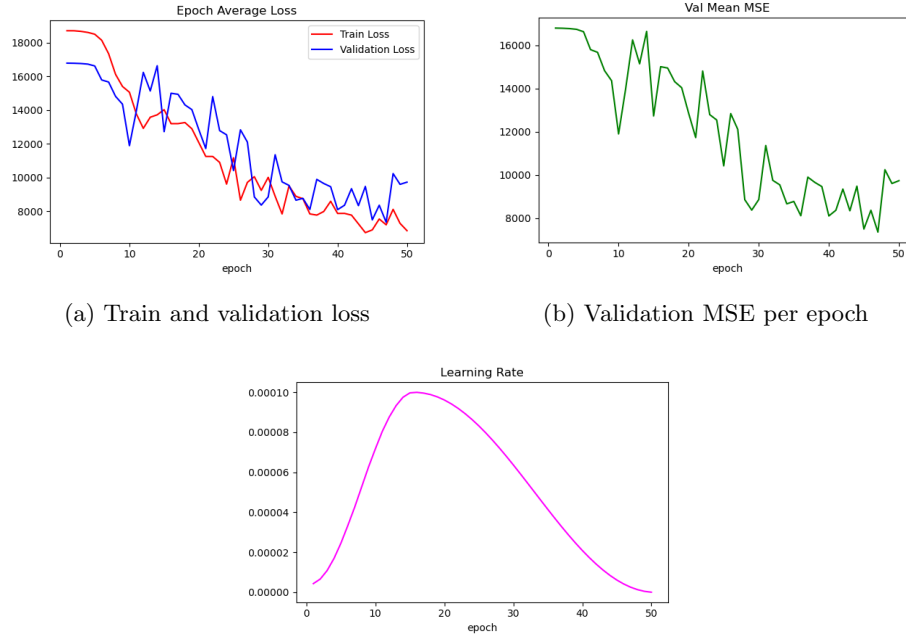


Fig. 1: Training results

References

1. “What is camera calibration? - matlab & simulink,” <https://www.mathworks.com/help/vision/ug/camera-calibration.html>, (Accessed on 08/30/2024).
2. “Camera calibration — camera calibration - youtube,” <https://www.youtube.com/watch?v=GUbWsXU1mac>, (Accessed on 08/30/2024).
3. K. Hata and S. Savarese, “Cs231a course notes 1: Camera models,” <https://web.stanford.edu/class/cs231a/course.notes/01-camera-models.pdf>, (Accessed on 08/30/2024).
4. T. Collins, “Contributions to Plane-based and Deformable 3D Reconstruction from Monocular Images,” Theses, Université Clermont Auvergne, Dec. 2021. [Online]. Available: <https://theses.hal.science/tel-04673567>
5. K. Liao, L. Nie, S. Huang, C. Lin, J. Zhang, Y. Zhao, M. Gabbouj, and D. Tao, “Deep learning for camera calibration and beyond: A survey,” 2024. [Online]. Available: <https://arxiv.org/abs/2303.10559>
6. A. Hagemann, M. Knorr, and C. Stiller, “Deep geometry-aware camera self-calibration from video,” in *International Conference on Computer Vision*, 2023, pp. 3438–3448.
7. B. Huang, J.-Q. Zheng, A. Nguyen, C. Xu, I. Gkouzionis, K. Vyas, D. Tuch, S. Giannarou, and D. S. Elson, “Self-supervised depth estimation in laparoscopic image using 3d geometric consistency,” 2023. [Online]. Available: <https://arxiv.org/abs/2208.08407>
8. Y. Hirohata, M. Sogabe, T. Miyazaki, T. Kawase, and K. Kawashima, “Confidence-aware self-supervised learning for dense monocular depth estimation in dynamic laparoscopic scene,” *Scientific Reports*, vol. 13, no. 1, 9 2023. [Online]. Available: <http://dx.doi.org/10.1038/s41598-023-42713-x>